

Dissolved Oxygen and Aquatic Invertebrate Abundance at North Campus Open Space

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Introduction

Wetland restoration projects are often evaluated using physical, chemical, and biological indicators because restoration success depends not only on habitat construction, but also on whether restored ecosystems support functioning biological communities. Aquatic invertebrates are useful indicators of wetland condition because their abundance and community composition can respond to changes in water quality, hydrology, vegetation, and habitat structure. Studies from multiple wetland systems show that restored or rehabilitated wetlands can support greater invertebrate abundance than imbalanced wetlands (Batzer et al. 2015); (Lu et al. 2020). This suggests that aquatic invertebrate communities can provide insight into restoration progress.

As a result, aquatic invertebrate monitoring especially relevant at North Campus Open Space (NCOS) because it is a restored wetland where long-term monitoring can help evaluate how biological communities respond to changing environmental conditions.

Dissolved oxygen (DO) is one important water-quality variable because aquatic organisms depend on oxygen for survival, metabolism, reproduction, and movement. Research has shown that low oxygen conditions can negatively affect aquatic invertebrates by reducing survival, emergence, growth, feeding, reproduction, and activity (Connolly, Crossland, and Pearson 2004); (Galic, Hawkins, and Forbes 2019); (Collins et al. 2019). However, the effects of DO are not always simple or uniform across taxa. Some invertebrates can tolerate moderate reductions in oxygen, while others are more sensitive to hypoxia or respond behaviorally by moving away from low-oxygen areas (Connolly, Crossland, and Pearson 2004); (Ekau et al. 2010). One study found that oxygen can be a strong predictor of macroinvertebrate richness, but other studies show that dissolved oxygen also interacts with factors such as temperature, water depth, precipitation, and habitat zone (Croijmans, De Jong, and Prins 2021); (Lucchesi, Chipps, and Schumann 2022). Therefore, dissolved oxygen may influence both the total abundance of aquatic invertebrates and the relative abundance of different taxa within the community.

Research conducted at other locations also suggests that dissolved oxygen can shape aquatic invertebrate community structure, not just total abundance. In Lake Huron coastal wetland wet meadows, dissolved oxygen gradients influenced oxygen-sensitive zoobenthos, although broader site and wetland differences also helped explain community patterns (Burton and Uzarski 2009). Similarly, one study found that seasonal hypoxia affected where macroinvertebrates were found in a reservoir, but taxa such as Cladocera and Copepoda were also influenced by water depth and monthly precipitation (Lucchesi, Chipps, and Schumann 2022). These findings suggest that, while dissolved oxygen is ecologically important, its relationship with invertebrate abundance may be impacted by site conditions, seasonal change, and taxon-specific tolerance.

At NCOS, previous analysis by UCSB students also examined relationships among algae, macroinvertebrates, and water quality. Their poster suggested that dissolved oxygen change over time and may be linked to macroinvertebrate abundance (Huynh, Russell, and Malone 2023). They also found that ostracods and copepods were among the most abundant taxa. However, their findings also indicated that other water-quality variables, such as conductivity, may influence diversity and abundance patterns. This is important because it indicates that dissolved oxygen levels should be interpreted as part of a broader set of environmental factors rather than as the only driver of invertebrate abundance.

Building on previous previous, our analysis examines the relationship between dissolved oxygen and aquatic invertebrate abundance at NCOS. Specifically, we ask: **What is the strength and direction of the relationship between dissolved oxygen and invertebrate abundance?** To answer this question, we analyzed median dissolved oxygen levels (mg/L) and invertebrate abundance for our five focus taxa: Ostracod, Copepod, Cladocera, Hemiptera Corixidae Boatman, and Nematode. Throughout our analysis, we examined the relationship

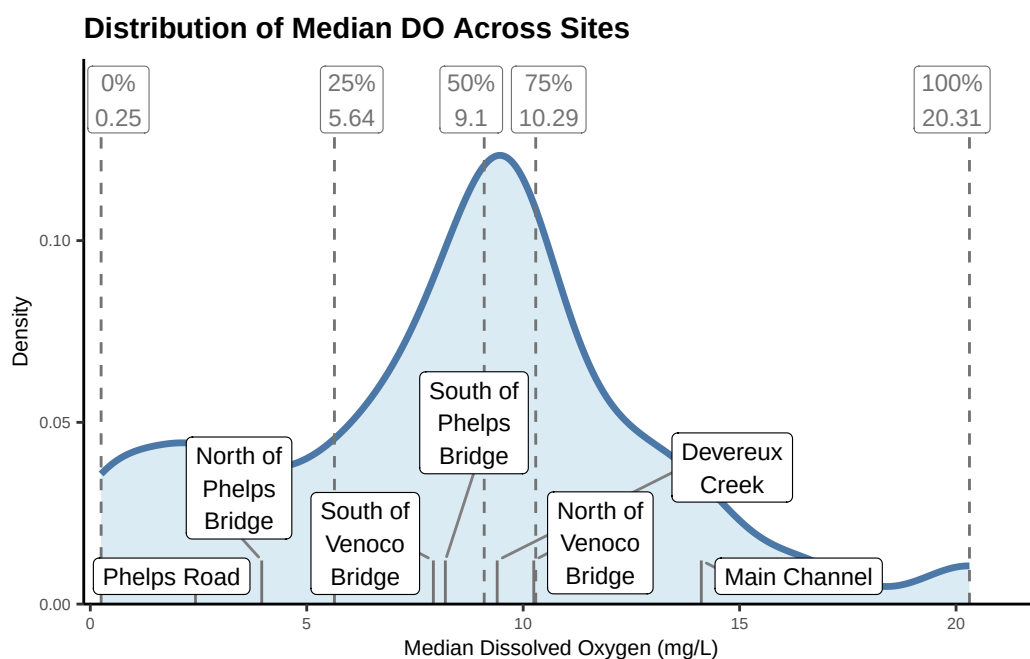
between DO and total invertebrate abundance, how DO varies across NCOS sites, taxon-specific abundance, abundance across DO categories, and community composition across low, medium, and high DO conditions.

Methods

Results

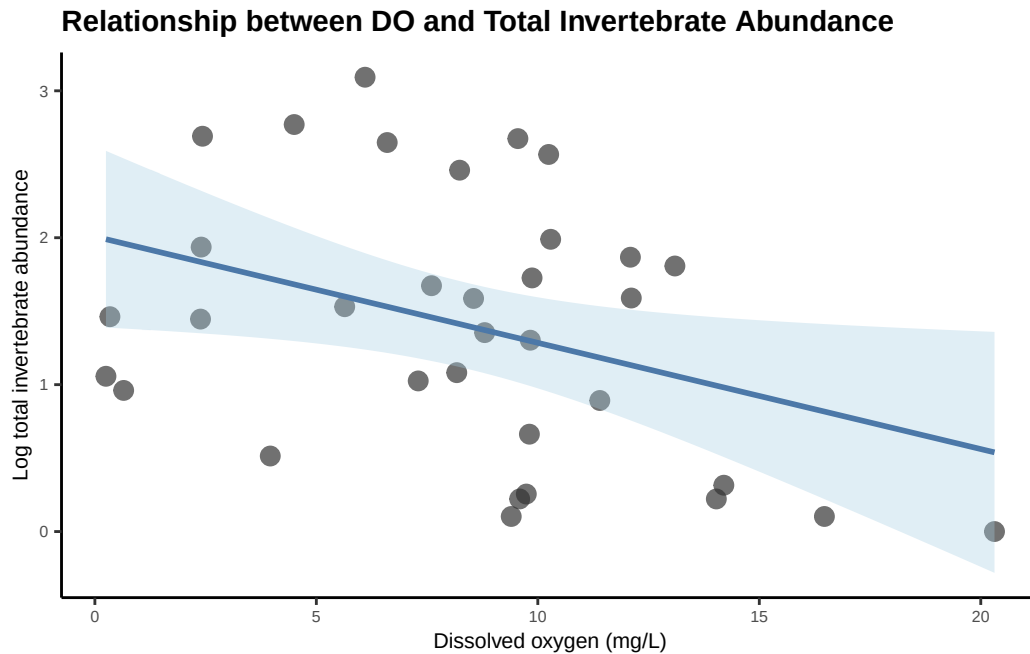
We first examined dissolved oxygen patterns across NCOS sampling sites to understand the oxygen conditions experienced by aquatic invertebrates. Median dissolved oxygen varied substantially among sites, indicating that invertebrates were not sampled under uniform oxygen conditions across the study area (Figure 1). Most median DO values were concentrated around approximately 8–11 mg/L, but the full range extended from very low values to values above 20 mg/L. Phelps Road and North of Phelps Bridge had relatively low median DO values, while Main Channel had the highest median DO values. This site-level variation provides important context for the main research question because it shows that the dataset includes a clear gradient in dissolved oxygen conditions.

Figure 1. Distribution of Median DO Across Sites



After establishing this DO gradient, we examined whether total invertebrate abundance was related to continuous dissolved oxygen values. Total abundance, summed across the five focus taxa and log-transformed, showed a weak-to-moderate negative relationship with median dissolved oxygen (Figure 2).

Figure 2. Relationship between DO and Total Invertebrate Abundance



A Pearson correlation test showed a statistically significant negative linear relationship between median DO and log-transformed total invertebrate abundance, $r = -0.375$, $t(32) = -2.29$, $p = 0.0289$. The 95% confidence interval for the correlation ranged from -0.633 to -0.042, which does not include zero, further supporting a negative relationship. This means that as dissolved oxygen increased, total invertebrate abundance tended to decrease. However, because the correlation was not close to -1, the relationship was not strong; dissolved oxygen explained some, but not all, of the variation in total abundance.

Pearson's product-moment correlation

```
data: total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
```

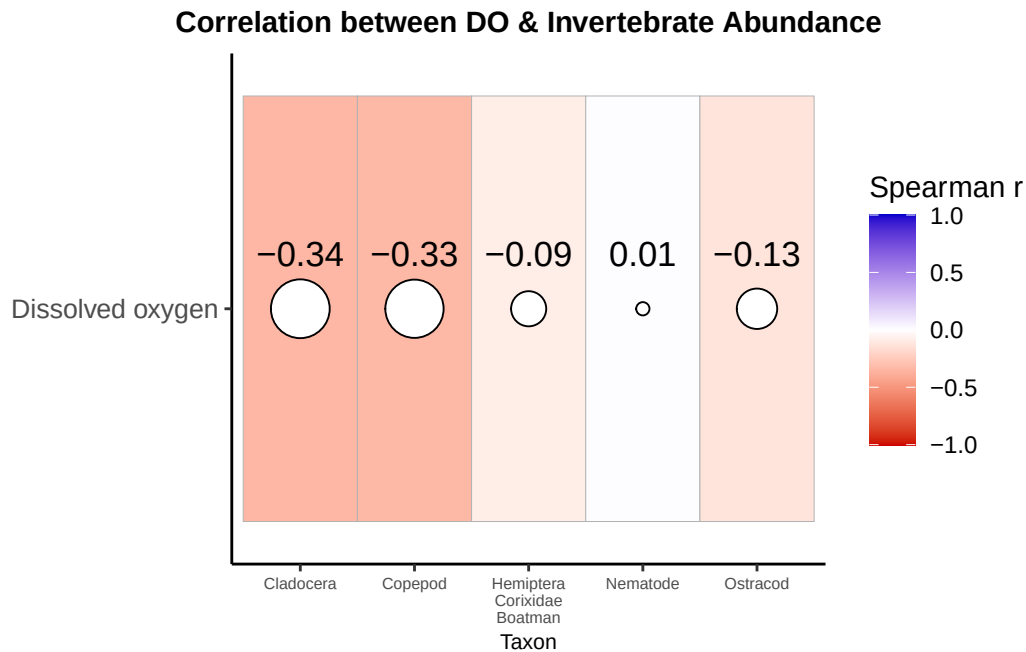
```

-0.63289820 -0.04217141
sample estimates:
      cor
-0.3749894

```

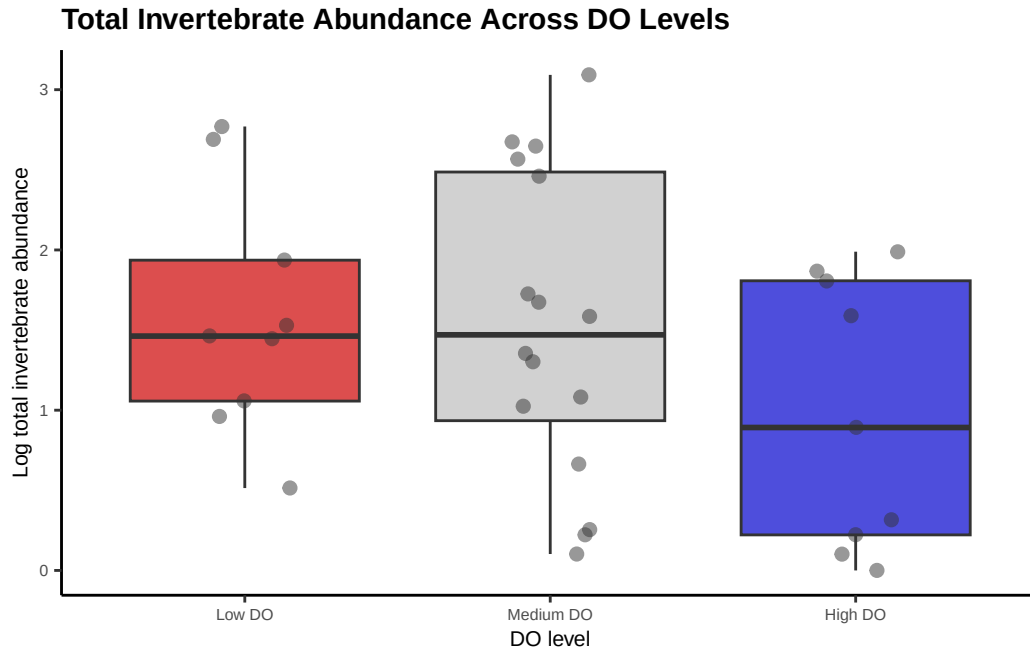
We then examined whether this relationship was consistent across individual taxa. Taxon-specific correlations showed that most focus taxa had weak negative relationships with dissolved oxygen, but the strength of the relationship differed among groups (Figure 3). Cladocera and Copepod had the strongest negative correlations with DO, while Ostracod and Hemiptera Corixidae Boatman had weaker negative correlations. Nematode showed almost no relationship with dissolved oxygen. This suggests that the community-level relationship between DO and abundance was not driven equally by all taxa, but instead reflected different taxon-specific patterns.

Figure 3: Correlation between DO & Invertebrate Abundance



To compare abundance across broader oxygen conditions, we grouped observations into low, medium, and high dissolved oxygen categories. Total log-transformed invertebrate abundance appeared lower in the High DO category than in the Low DO and Medium DO categories (Figure 4). However, there was substantial overlap among the categories, especially in the Medium DO group.

Figure 4: Total Invertebrate Abundance Across DO Levels



A Kruskal-Wallis test showed that total abundance did not differ significantly across the three DO categories, $X^2 = 2.00$, $df = 2$, $p = 0.368$. This indicates that, although the visualization suggests lower abundance under high DO conditions, there was not enough statistical evidence to conclude that total abundance differed significantly among the DO categories.

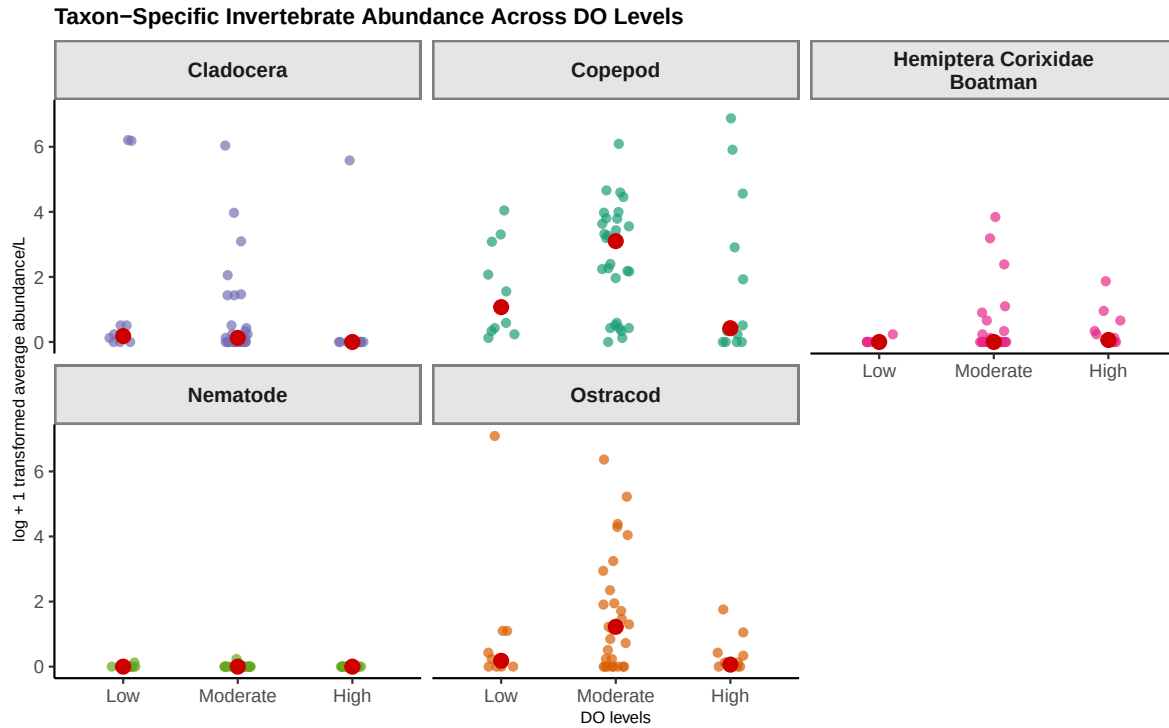
Kruskal-Wallis rank sum test

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data: log_total_abundance by do_category
```

```
Kruskal-Wallis chi-squared = 2.0001, df = 2, p-value = 0.3679
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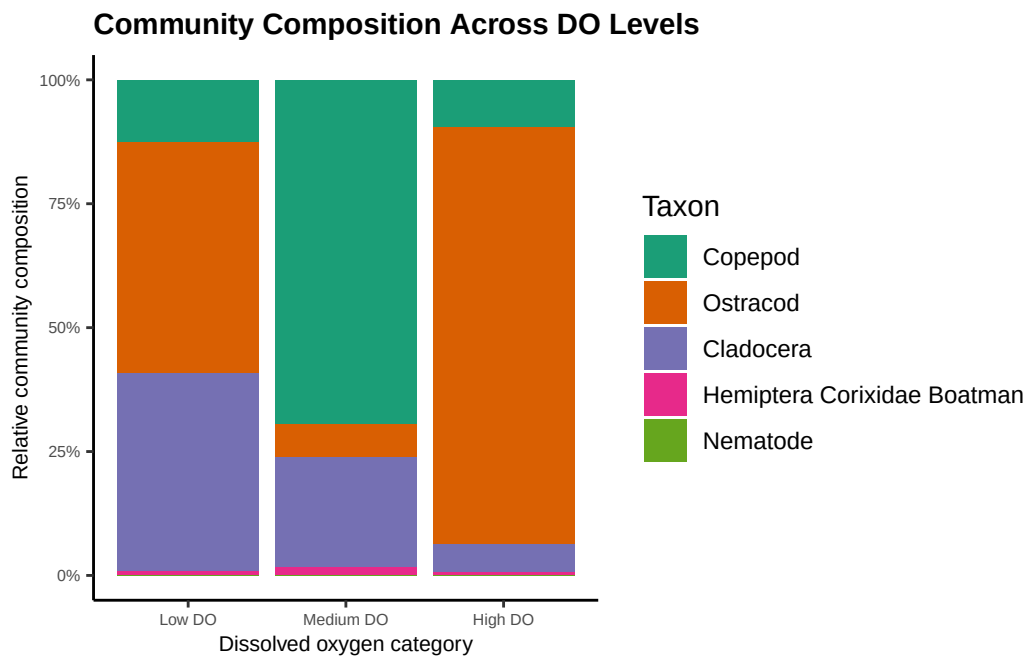
Taxon-specific abundance patterns across DO categories showed more detail than the community-level comparison. Copepods and Ostracods had higher and more variable abundances in the Moderate DO category, while Cladocera showed some high-abundance observations in the Low and Moderate DO categories (Figure 5). Nematodes remained close to zero across all dissolved oxygen levels, and Hemiptera Corixidae Boatman generally had low abundance, with only a few higher observations in Moderate DO. These patterns suggest that dissolved oxygen may be related to abundance differently depending on the taxon, rather than producing one consistent response across the whole invertebrate community.

Figure 5. Taxon-Specific Invertebrate Abundance Across DO Levels



Finally, we examined whether the relative composition of the invertebrate community changed across dissolved oxygen categories. Community composition shifted clearly among Low, Medium, and High DO groups (Figure 6). Low DO sites had a more mixed community dominated mostly by Ostracods and Cladocera. Moderate DO sites were dominated by Copepods and Ostracods, with smaller contributions from Cladocera and Hemiptera Corixidae Boatman. High DO sites were dominated primarily by Copepods, with Cladocera making up a smaller portion of the community and Ostracods contributing very little. Hemiptera Corixidae Boatman and Nematodes made up small proportions of the community across all DO categories.

Figure 6. Community Composition Across DO Levels



A Pearson's chi-squared test showed that taxon composition differed significantly across DO bins, $X^2 = 2859.2$, $df = 8$, $p < 2.2e-16$. This provides strong evidence that the relative abundances of the focus taxa were not evenly distributed across low, medium, and high DO conditions. In other words, dissolved oxygen categories were strongly associated with changes in community composition, even though total abundance did not differ significantly across DO categories.

	Cladocera	Copepod	Hemiptera Corixidae Boatman	Nematode	Ostracod
low	980.2667	115.8667	0.2666667	0.1333333	1202.933333
moderate	510.6667	1170.4000	83.8666667	0.2666667	1057.866667
high	264.2667	1453.4667	8.8000000	0.0000000	7.866667

Pearson's Chi-squared test

data: chi_square_table
X-squared = 2859.2, df = 8, p-value < 2.2e-16

Overall, the results suggest that the relationship between dissolved oxygen and invertebrate abundance at NCOS was weak-to-moderate and generally negative. Total invertebrate abun-

dance decreased as DO increased when DO was analyzed as a continuous variable, but abundance differences across broad DO categories were not statistically significant. The strongest evidence of a DO-related pattern was seen in community composition, where the relative abundance of different taxa shifted strongly across oxygen categories. This indicates that dissolved oxygen may be more strongly related to which taxa dominate the invertebrate community than to total abundance alone.

Discussion

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